Perfect Secrecy, the One-Time Pad, and Computational Security

CS 171: Cryptography

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Overview

Defining Secure Encryption: Formally

Definition 1 (Perfect Secrecy)

An encryption scheme $(\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$ with message space $\mathcal{M}$ is **perfectly secret** if for every probability distribution over $\mathcal{M}$, every message $m \in \mathcal{M}$, and every ciphertext c with $\Pr[C = c] > 0$:

$$\Pr[M = m \mid C = c] = \Pr[M = m].$$

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$$\Pr[M = m \mid C = c] = \Pr[M = m].$$

Equivalent formulation

Or, equivalently, if for every two messages $m, m' \in \mathcal{M}$ and every ciphertext c (in the ciphertext space):

$$\Pr[\mathsf{Enc}_K(m) = c] = \Pr[\mathsf{Enc}_K(m') = c].$$

Definition 3 (Game Style)

Game $\mathsf{PrivK}^{\mathsf{eav}}_{\mathcal{A},\Pi}$

1. $\mathcal{A}$ outputs $m_0, m_1 \in \mathcal{M}$.
2. $b \leftarrow \{0,1\}$, $k \leftarrow \mathsf{Gen}()$, $c \leftarrow \mathsf{Enc}_k(m_b)$.
3. Challenge ciphertext c is given to $\mathcal{A}$.
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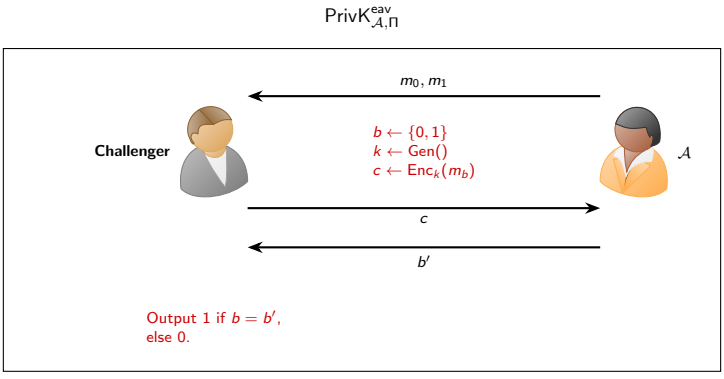
Definition 2 (Perfect Indistinguishability)

Encryption scheme $\Pi = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$ with message space $\mathcal{M}$ is **perfectly indistinguishable** if $\forall\, \mathcal{A}$ it holds that

$$\Pr\left[\mathsf{PrivK}^{\mathsf{eav}}_{\mathcal{A},\Pi} = 1\right] = \tfrac{1}{2}.$$

- *eav* is for *eavesdropper*.

$\mathsf{PrivK}^{\mathsf{eav}}_{\mathcal{A},\Pi}$: Challenger vs. Adversary



Remark

$\mathcal{A}$ can always succeed with probability $\frac{1}{2}$ by guessing b' at random; security requires it can do no better.

Two Equivalent Definitions

Lemma (prove on your own)

An encryption scheme is **perfectly secret** if and only if it is **perfectly indistinguishable**.

- The two definitions capture the same intuition: the ciphertext reveals nothing about which message was encrypted.
- The game-based (indistinguishability) formulation will be the template for all later security definitions.

The One-Time Pad

Definition 3 (One-Time Pad)

Fix an integer ℓ, then let $\mathcal{M} = \mathcal{K} = \mathcal{C} = \{0,1\}^\ell$:

- Gen: output a uniform value from $\mathcal{K}$.
- $\mathsf{Enc}_k(m)$: where $m \in \{0,1\}^\ell$, output $c := k \oplus m$.
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Security (perfect secrecy)

$\forall\, m, c : \Pr[\mathsf{Enc}_K(m) = c] = 2^{-\ell}$. Or,
$\forall\, m, m', c : \Pr[\mathsf{Enc}_K(m) = c] = \Pr[\mathsf{Enc}_K(m') = c]$.

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- Not used anymore, why not?
  - The key is **as long as the message**.
  - Can't reuse the key.
  - Broken under a **known-plaintext attack**.

Key reuse is fatal

If $c_1 = k \oplus m_1$ and $c_2 = k \oplus m_2$, then $c_1 \oplus c_2 = m_1 \oplus m_2$, leaking information about the messages.

Can We Make $|\mathcal{M}| > |\mathcal{K}|$?

The question

The one-time pad needs a key as long as the message ($|\mathcal{K}| = |\mathcal{M}|$). Can a *perfectly secret* scheme use a key space smaller than its message space, i.e. $|\mathcal{M}| > |\mathcal{K}|$?

- Intuitively: more messages than keys means some ciphertext cannot "cover" every message, which should break perfect secrecy.
- The next theorem shows the answer is **no**.

Optimality of the One-Time Pad

Theorem 4 (Shannon)

If $\Pi = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$ *is a perfectly secret encryption scheme with message space* $\mathcal{M}$ *and key space* $\mathcal{K}$*, then* $|\mathcal{M}| \leq |\mathcal{K}|$.

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- Then $|\mathcal{M}(c)| \leq |\mathcal{K}| < |\mathcal{M}|$, so $\exists\, m' \in \mathcal{M}$ with $m' \notin \mathcal{M}(c)$.
- For such m': $\Pr[M = m' \mid C = c] = 0 \neq \Pr[M = m']$, contradicting perfect secrecy. □

Computational Security

- A **relaxation** of perfect security:
  - Security only against **efficient** adversaries.
  - Security can fail with some **very small** probability.
- Two approaches:
  - **Concrete** security.
  - **Asymptotic** security.

Why relax?

This relaxation is what allows schemes with short keys (e.g. keys much shorter than the messages) to be considered secure — escaping Shannon's bound.

Concrete Security

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- 2^{80} is roughly the computation that can be performed by super-computers in one year or so.
- 2^{-60} is roughly the probability that an event happens once every 100 billion years.

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- However, it doesn't yield a clean theory:
  - Depends on the computational model.
  - Need to change schemes as (t, ϵ) need to be updated.
- Need schemes that allow **tuning** (t, ϵ) as desired.

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  - t and ϵ are now functions of n:
  - $t \rightarrow$ probabilistic polynomial time (**PPT**) in n.
  - $\epsilon \rightarrow$ a **negligible** function in n.

Polynomial and Negligible

Polynomial

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Negligible

A function $f : \mathbb{Z}^+ \to [0, 1]$ is **negligible** if for every polynomial p it holds that for all large enough n:

$$f(n) < \frac{1}{p(n)}.$$

- Typical example: $f(n) = \text{poly}(n) \cdot 2^{-\alpha n}$.

Negligible Function (Formally)

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A function $f : \mathbb{Z}^+ \to [0, 1]$ is **negligible** if for every polynomial p,

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Example 8

Prove that $f(n) = 2^{-n}$ is a negligible function.

- For any polynomial p, eventually $2^{-n} < 1/p(n)$ since 2^n grows faster than every polynomial.

Is This a Negligible Function?

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- $f(n) = \begin{cases} 2^{-n} & \text{for } n \bmod 2 = 0 \\ n^{-c} & \text{for } n \bmod 2 = 1 \end{cases}$ (for some constant c)?

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- $f(n) = \begin{cases} 2^{-n} & \text{for } n \bmod 2 = 0 \\ n^{-c} & \text{for } n \bmod 2 = 1 \end{cases}$ (for some constant c)?
  - **No** — for every odd n, $f(n) = n^{-c}$ is *not* below $1/p(n)$ for $p(n) = n^c$; the bound must hold for *all* large enough n, not just the even ones.

Choice of Polynomial and Negligible

- Using PPT for "efficient" machines is borrowed from **complexity theory**.
- Also some nice **closure properties**:
  - $\text{poly}(n) \cdot \text{poly}(n)$ is still $\text{poly}(n)$.
  - $\text{poly}(n) \cdot \text{negl}(n)$ is still $\text{negl}(n)$.

Why these closures matter

They let us compose efficient algorithms (still efficient) and union-bound over polynomially many negligible events (still negligible) — the workhorses of security proofs.

Concrete vs. Asymptotic

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Asymptotic

A scheme is **secure** if any PPT adversary succeeds in breaking the scheme with at most **negligible** probability.

$$\text{fixed } t, \epsilon$$

$$\Downarrow$$

$$t = \text{poly}(n), \quad \epsilon = \text{negl}(n)$$

Defining Computationally Secure Encryption (Syntax)

Definition 9 (Private-Key Encryption Scheme)

A **private-key encryption scheme** is a tuple of algorithms $(\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$:

- $\mathsf{Gen}(1^n)$: outputs a key k (assume $|k| > n$).
- $\mathsf{Enc}_k(m)$: takes key k and message $m \in \{0,1\}^*$ as input; outputs ciphertext $c \leftarrow \mathsf{Enc}_k(m)$.
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Correctness

For all n, all k output by $\mathsf{Gen}(1^n)$, and all $m \in \{0,1\}^*$:

$$\mathsf{Dec}_k(\mathsf{Enc}_k(m)) = m.$$

Computational Indistinguishability

Game $\mathsf{PrivK}^{\mathsf{eav}}_{\mathcal{A},\Pi}(n)$

1. $\mathcal{A}$ (given 1^n) outputs $m_0, m_1 \in \{0,1\}^*$ with $|m_0| = |m_1|$.
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Definition 10 (Computational Indistinguishability (EAV-security))

Π is **computationally indistinguishable** if for every **PPT** $\mathcal{A}$:
$\Pr[\mathsf{PrivK}^{\mathsf{eav}}_{\mathcal{A},\Pi}(n) = 1] \leq \frac{1}{2} + \mathsf{negl}(n)$.

Caveat

Does **not** hide message length!

Distinguishing Variant

Game $\mathsf{PrivK}^{\mathsf{eav}}_{\mathcal{A},\Pi}(n,d)$

1. $\mathcal{A}$ (given 1^n) outputs $m_0, m_1 \in \{0,1\}^*$ with $|m_0| = |m_1|$.
2. $b = d$, $k \leftarrow \mathsf{Gen}(1^n)$, $c \leftarrow \mathsf{Enc}_k(m_b)$.
3. Challenge ciphertext c is given to $\mathcal{A}$.
4. $\mathcal{A}$ outputs b'. The output of $\mathcal{A}$ is $\mathsf{out}_{\mathcal{A}}\big(\mathsf{PrivK}^{\mathsf{eav}}_{\mathcal{A},\Pi}(1^n,d)\big)$.

- Here, $\mathsf{PrivK}^{\mathsf{eav}}_{\mathcal{A},\Pi}(1^n,d)$ is the same as $\mathsf{PrivK}^{\mathsf{eav}}_{\mathcal{A},\Pi}(1^n)$ except that we set $b = d$.

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Definition 11 (Indistinguishability, distinguishing form)

Π is computationally indistinguishable if for every PPT $\mathcal{A}$:

$$\Big|\Pr\big[\mathsf{out}_{\mathcal{A}}\big(\mathsf{PrivK}^{\mathsf{eav}}_{\mathcal{A},\Pi}(1^n,1)\big)=1\big] - \Pr\big[\mathsf{out}_{\mathcal{A}}\big(\mathsf{PrivK}^{\mathsf{eav}}_{\mathcal{A},\Pi}(1^n,0)\big)=1\big]\Big| \leq \mathsf{negl}(n).$$

Summary

- **Perfect secrecy** $\equiv$ **perfect indistinguishability** (game $\mathsf{PrivK}^{\mathsf{eav}}_{\mathcal{A},\Pi}$, winning prob. exactly $\frac{1}{2}$).
- **One-time pad**: $c = k \oplus m$ — perfectly secret, but key as long as message and not reusable.
- **Shannon's bound**: perfect secrecy requires $|\mathcal{K}| \geq |\mathcal{M}|$.
- **Computational security** relaxes this: efficient (PPT) adversaries, negligible failure.
  - **Concrete**: (t, ϵ)-security.
  - **Asymptotic**: security parameter n, $\mathsf{poly}(n)$ time, $\mathsf{negl}(n)$ advantage.
- **EAV-security**: $\Pr[\mathsf{PrivK}^{\mathsf{eav}}_{\mathcal{A},\Pi}(n) = 1] \leq \frac{1}{2} + \mathsf{negl}(n)$ for all PPT $\mathcal{A}$.